BSc CS IT — PRACTICAL FILE

PRACTICAL / TUTORIAL NO. 9

BLACK BOX & WHITE BOX TESTING

Subject: Software Testing

Student Name	RUTVIK VALA
Enrollment Number	2504030400048
Program	BSc CS IT

Includes workbook worksheets and the four screenshot images supplied for this practical.

Implementation

Tutorial 9 — Black-Box & White-Box Testing | Implementation Details	Unnamed: 1
Parameter	Details
Project Name	Student Login & Registration System
Application/Software	Sample Web Application
Module Name	Login and Registration
SDLC Model	Waterfall Model (academic example)
Programming Language	HTML, CSS, JavaScript
IDE/Tool Used	Visual Studio Code, Web Browser
Input/Test Data	Valid credentials, wrong password, blank fields
Student Name	Rutvik Vala
Enrollment No.	2504030400048
Aim	Analyze Black-Box, White-Box, Integration and System Testing.

Test Case Details

Test Case Details	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6	Unnamed: 7
Test Case ID	Requirement ID	Module	Feature Under Test	Scenario	Objective	Test Type	Technique
TC-001	REQ-LOGIN-01	Login	Authentication	Valid Login	Verify correct credentials	Functional	Black-Box
TC-002	REQ-LOGIN-01	Login	Authentication	Invalid Password	Verify wrong password is rejected	Functional	Black-Box
TC-003	REQ-LOGIN-02	Login	Form Validation	Empty Fields	Verify required-fiel d validation	Functional	Black-Box
TC-004	REQ-REG-01	Registration	Form Validation	Invalid Email	Verify invalid email is rejected	Functional	Black-Box

Test Execution

Test Case Execution	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6
Step No.	Test Case ID	Test Steps	Input	Expected Result	Actual Result	Status
1	TC-001	Open application	Application URL	Login page opens	Login page opens	Pass
2	TC-001	Enter valid credentials	student / 1234	Credentials accepted	Credentials accepted	Pass
3	TC-001	Click Login	Login button	Login Successful	Login Successful	Pass
1	TC-002	Enter wrong password	student / wrong	Invalid Credentials	Invalid Credentials	Pass
1	TC-003	Leave fields blank	blank / blank	Fields are required	Fields are required	Pass
1	TC-004	Enter invalid email	student@	Validation message	Validation message	Pass

Testing Activities

Testing Activities Performed	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4
Sr. No.	SDLC Phase	STLC Phase	Testing Activity	Deliverable / Artifact
1	Requirement Analysis	Requirement Analysis	Identify requirements	Requirement Document
2	Design	Test Planning	Prepare test plan	Test Plan
3	Design	Test Case Development	Create test cases	Test Case Document
4	Development	Environment Setup	Prepare application and browser	Test Environment
5	Testing	Test Execution	Execute test cases	Test Execution Report

Execution Report

Test Execution Report	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6
Test Case ID	Module Name	Test Scenario	Expected Result	Actual Result	Status	Remarks
TC-001	Login	Valid Login	Login Successful	Login Successful	Pass	Working correctly
TC-002	Login	Invalid Password	Invalid Credentials	Invalid Credentials	Pass	Wrong password rejected
TC-003	Login	Empty Fields	Fields are required	Fields are required	Pass	Validation working
TC-004	Registration	Invalid Email	Validation message	Validation message	Pass	Validation working

Summary Report

Test Summary Report	Unnamed: 1
Metric	Value
Total Test Cases Designed	
Total Test Cases Executed	
Total Passed	
Total Failed	
Total Blocked	0
Total Not Executed	0
Pass Percentage	
Fail Percentage	
Overall Result	
Tester Name	Student QA
Execution Date	28-10-2026

MCQ Answers

Tutorial 9 — MCQ Answers	Unnamed: 1	Unnamed: 2
Q. No.	Correct Answer	Explanation
1	C) Black-Box Testing	Checks functionality without examining source code.
2	D) White-Box Testing	Requires knowledge of internal program code.
3	B) Communication between modules	Integration testing checks module interactions.
4	C) Complete Application Behavior	System testing validates the complete application.
5	B) White-Box Testing	Checks execution paths and branch coverage.

Viva Questions

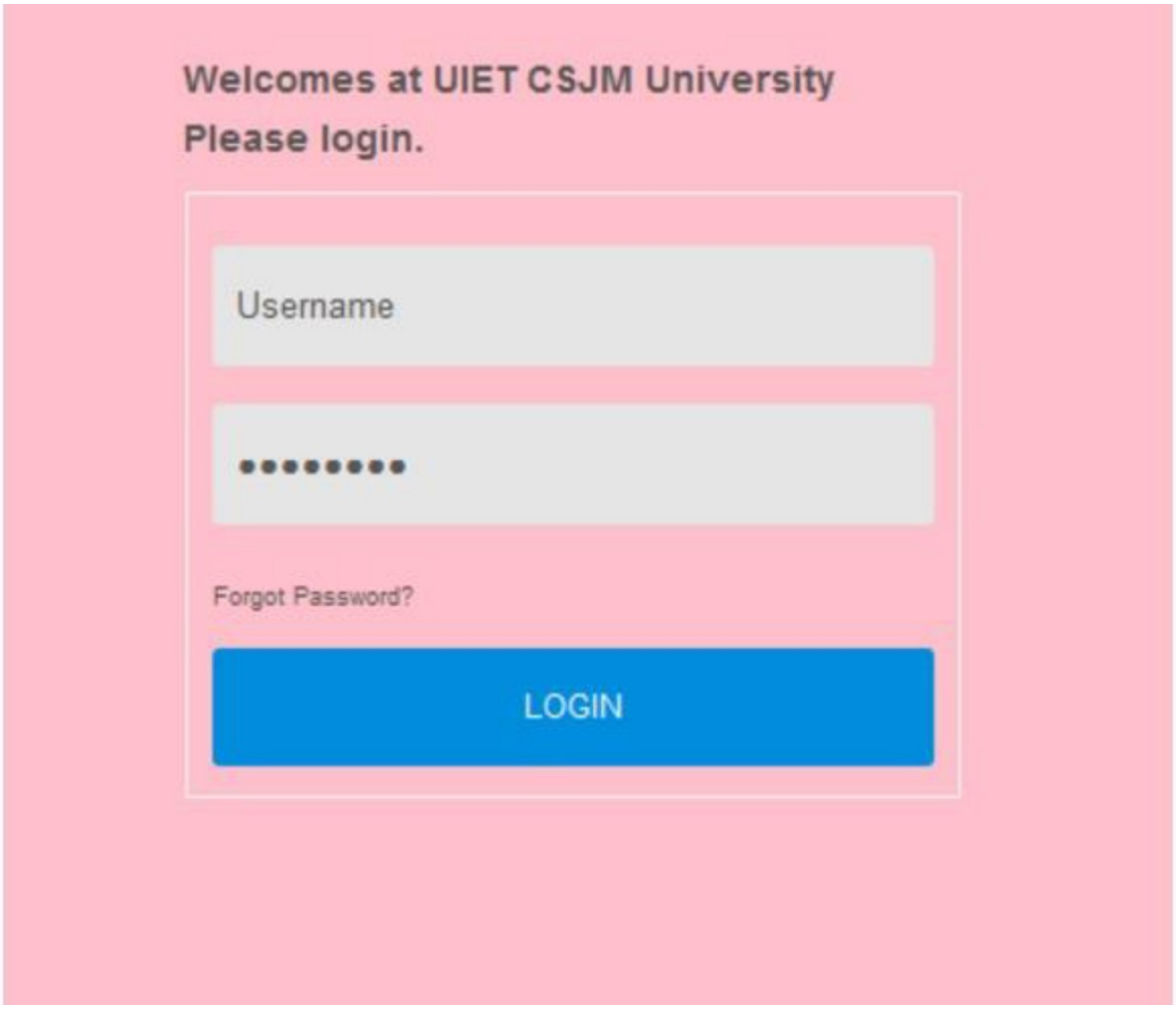
Viva Questions	Unnamed: 1
Question	Answer
What is Black-Box Testing?	Testing functionality without examining internal source code.
What is White-Box Testing?	Testing internal code, logic, paths, and branches.
What is Integration Testing?	Testing communication between integrated modules.
What is System Testing?	Testing the complete application as a whole.
What is a test case?	A set of inputs, steps, and expected results used to test a feature.

Screenshots

Tutorial 9 — Illustrative Output Screenshots
Note: These are illustrative mockups, not screenshots captured from a live application.
Output 1: 01_login_valid.png
Output 2: 02_login_invalid.png
Output 3: 03_empty_fields.png
Output 4: 04_test_summary.png

SCREENSHOT EVIDENCE

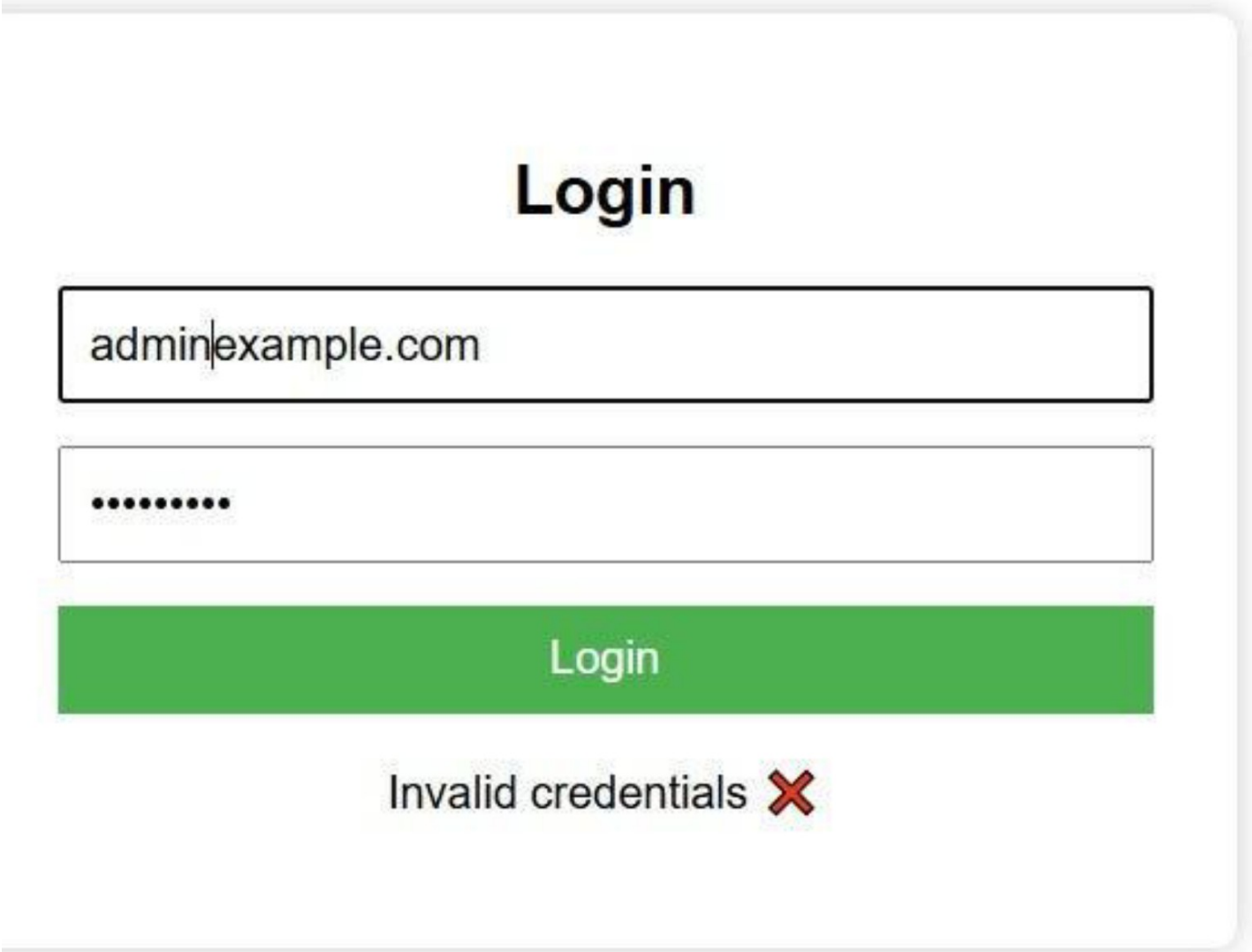
Screenshot 1 — Login Page (University Login Example)



Screenshot supplied by the student for practical reference/evidence.

SCREENSHOT EVIDENCE

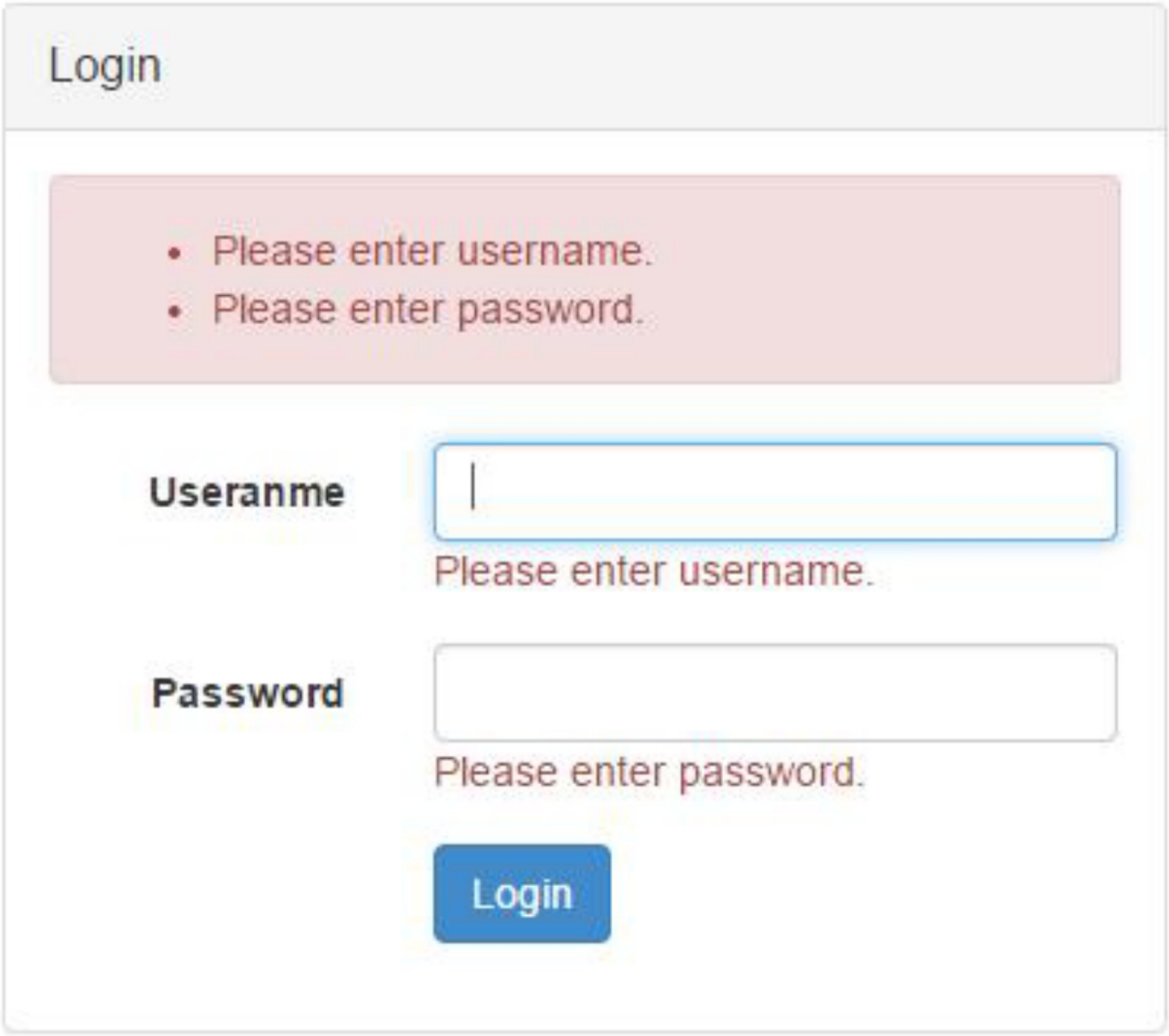
Screenshot 2 — Login with Invalid Credentials



Screenshot supplied by the student for practical reference/evidence.

SCREENSHOT EVIDENCE

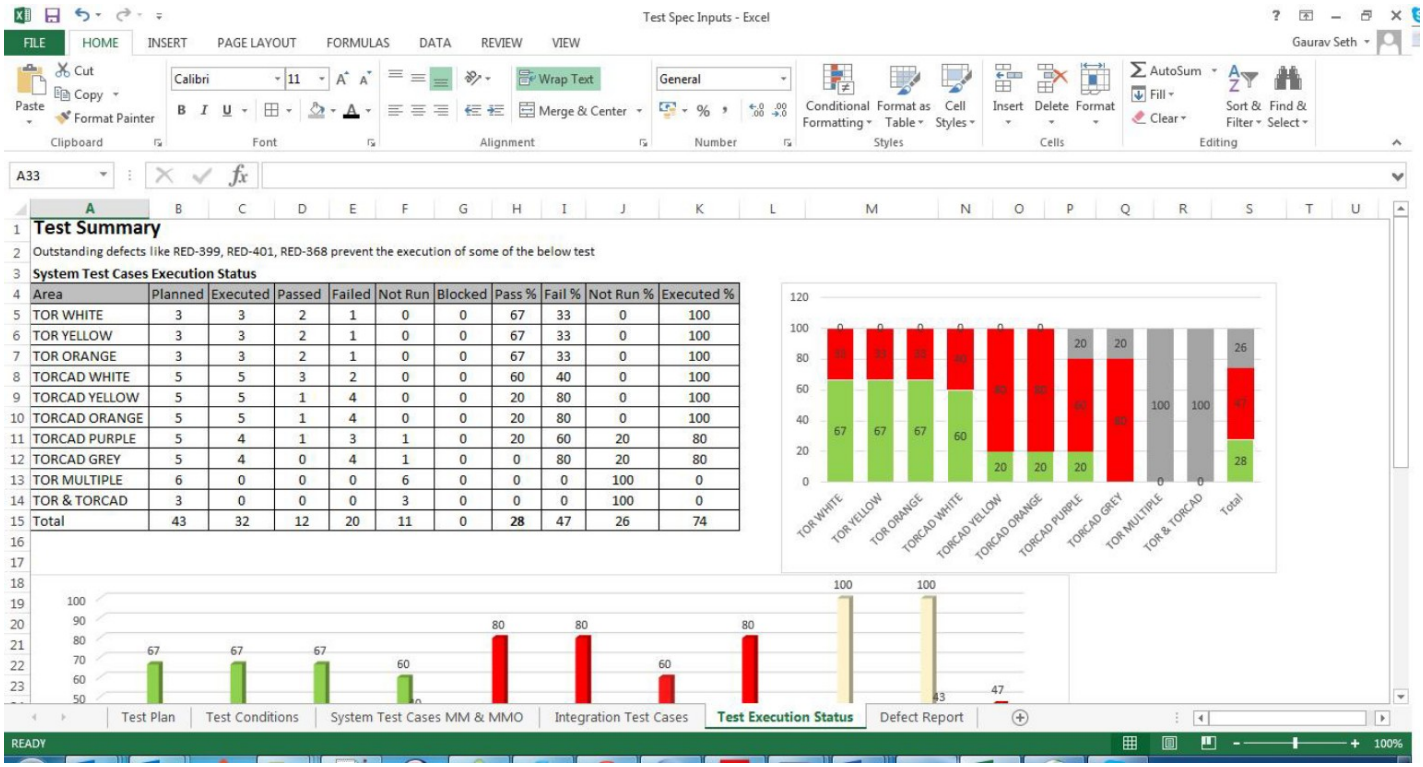
Screenshot 3 — Empty Login Form Validation



Screenshot supplied by the student for practical reference/evidence.

SCREENSHOT EVIDENCE

Screenshot 4 — Test Execution Status / Excel Summary



Area	Planned	Executed	Passed	Failed	Not Run	Blocked	Pass %	Fail %	Not Run %	Executed %
TOR WHITE	3	3	2	1	0	0	67	33	0	100
TOR YELLOW	3	3	2	1	0	0	67	33	0	100
TOR ORANGE	3	3	2	1	0	0	67	33	0	100
TORCAD WHITE	5	5	3	2	0	0	60	40	0	100
TORCAD YELLOW	5	5	1	4	0	0	20	80	0	100
TORCAD ORANGE	5	5	1	4	0	0	20	80	0	100
TORCAD PURPLE	5	4	1	3	1	0	20	60	20	80
TORCAD GREY	5	4	0	4	1	0	0	80	20	80
TOR MULTIPLE	6	0	0	0	6	0	0	0	100	0
TOR & TORCAD	3	0	0	0	3	0	0	0	100	0
Total	43	32	12	20	11	0	28	47	26	74

Screenshot supplied by the student for practical reference/evidence.